

# Variation of Parameters



## The Wronskian

- 1 For  $y_1 = e^x$ ,  $y_2 = e^{2x}$  (solutions of  $y'' - 3y' + 2y = 0$ ), compute the Wronskian  $W = y_1 y_2' - y_1' y_2$ .
  - 2 For  $y_1 = \cos x$ ,  $y_2 = \sin x$ , compute  $W$ .
  - 3 For  $y_1 = e^{3x}$ ,  $y_2 = x e^{3x}$  (solutions of a repeated-root equation), compute  $W$ .
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## The variation of parameters formula

- 4 State the variation of parameters formula for a particular solution of  $y'' + p(x)y' + q(x)y = g(x)$ , given independent solutions  $y_1, y_2$  of the homogeneous equation with Wronskian  $W$ .
  - 5 Explain why the ODE must first be written in standard form  $y'' + p(x)y' + q(x)y = g(x)$  (leading coefficient 1) before applying the variation of parameters formula.
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## A forcing term with no undetermined-coefficients trial form

- 6 Solve  $y'' + y = \tan x$  for  $-\frac{\pi}{2} < x < \frac{\pi}{2}$  using variation of parameters.
  - 7 Solve  $y'' + y = \sec x$  for  $-\frac{\pi}{2} < x < \frac{\pi}{2}$  using variation of parameters.
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## A repeated root, with forcing dividing by $x$

- 8 Solve  $y'' - 2y' + y = \frac{e^x}{x}$  ( $x > 0$ ) using variation of parameters.
  - 9 Solve  $y'' - 4y' + 4y = \frac{e^{2x}}{x^2}$  ( $x > 0$ ) using variation of parameters.
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## Comparing with undetermined coefficients

- 10 Solve  $y'' - y = e^{2x}$  by variation of parameters, then check the particular solution using undetermined coefficients.
- 11 Solve  $y'' - y = 4$  (a constant forcing) by variation of parameters, and check against the obvious undetermined-coefficients guess  $y_p = A$ .

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## When to prefer each method

- 12 For  $y'' + 4y = x^3 - 2x$ , which method is more efficient: undetermined coefficients or variation of parameters? Explain.
- 13 For  $y'' + y = \csc x$  (for  $0 < x < \pi$ ), which method must be used, and why?

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## Solving $y'' + y = \csc x$ in full

- 14 Complete the solution of  $y'' + y = \csc x$  ( $0 < x < \pi$ ) started above, using  $y_1 = \cos x$ ,  $y_2 = \sin x$ ,  $W = 1$ .
- 15 Verify (by checking the coefficient of the leading term only) that  $y_p = -x \cos x$  contributes a term of the correct order when substituted into  $y'' + y$ .

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## Variation of parameters with a shifted independent variable

- 16 For the Cauchy–Euler-type substitution used in some ODE courses,  $y_1 = x$ ,  $y_2 = x^{-1}$  solve  $x^2 y'' + xy' - y = 0$  for  $x > 0$ . Rewrite this equation in the standard form  $y'' + p(x)y' + q(x)y = g(x)$  needed for variation of parameters when the right side is  $x^2 \cdot 3x = 3x^3$  (i.e., solve  $x^2 y'' + xy' - y = 3x^3$ ).
- 17 Continue the problem above: with  $y_1 = x$ ,  $y_2 = x^{-1}$ , compute  $W = y_1 y_2' - y_1' y_2$ .
- 18 Finish solving  $x^2 y'' + xy' - y = 3x^3$  ( $x > 0$ ) using  $y_1 = x$ ,  $y_2 = x^{-1}$ ,  $W = -2x^{-1}$ , and  $g(x) = 3x$  (from the two questions above).
- 19 Check the particular solution  $y_p = \frac{3}{8}x^3$  found above by substituting directly into  $x^2 y'' + xy' - y = 3x^3$ .

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## A summary comparison

- 20 List two advantages of variation of parameters over undetermined coefficients.
- 21 List one advantage of undetermined coefficients over variation of parameters, when it applies.
- 22 True or false, with a one-sentence justification: "Variation of parameters requires knowing the general solution of the associated homogeneous equation."
- 23 For  $y'' + 4y = 8 \tan 2x$  ( $-\frac{\pi}{4} < x < \frac{\pi}{4}$ ), which method must be used, and what are  $y_1$ ,  $y_2$ , and  $W$ ?

- 24** In one sentence, summarize when a student should reach for variation of parameters instead of undetermined coefficients.